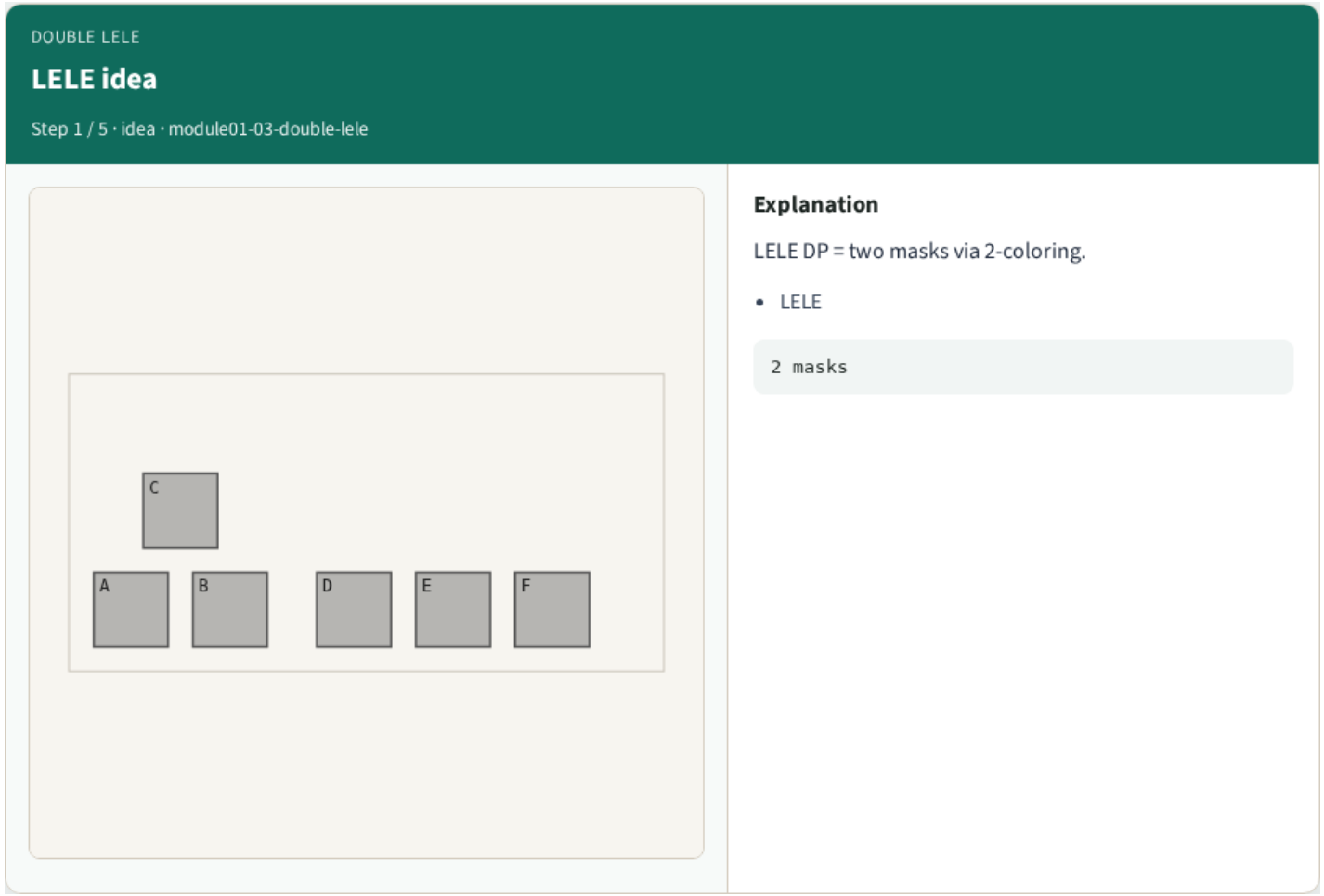
LELE overview

LELE double patterning assigns two lithography masks by 2-coloring conflicts

The idea

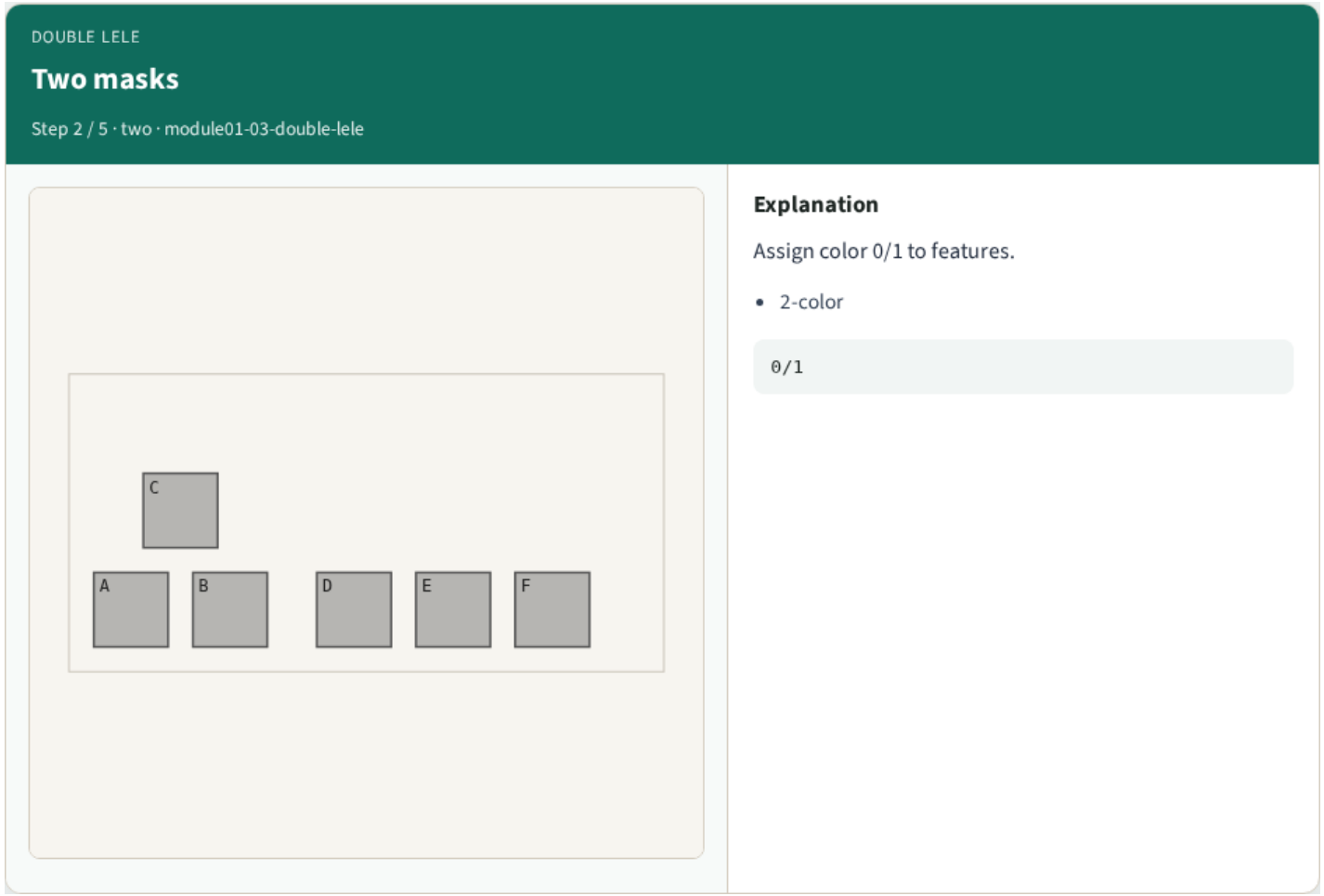
- GOLDEN: lele_double ok=False on the full graph (triangle A-B-C)

LELE idea



LELE idea

Two masks



Two masks

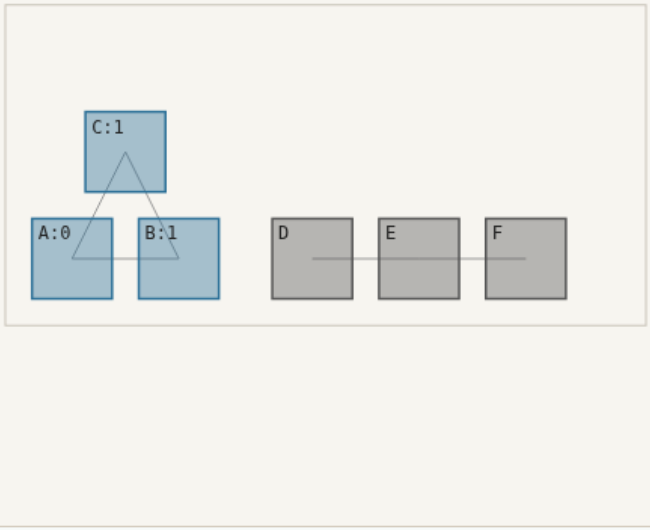
Color attempt

DOUBLE LELE

Color attempt

Step 3 / 5 · try · module01-03-double-lele

ok=false



Explanation

BFS 2-color the conflict graph.

- BFS

```
attempt
```

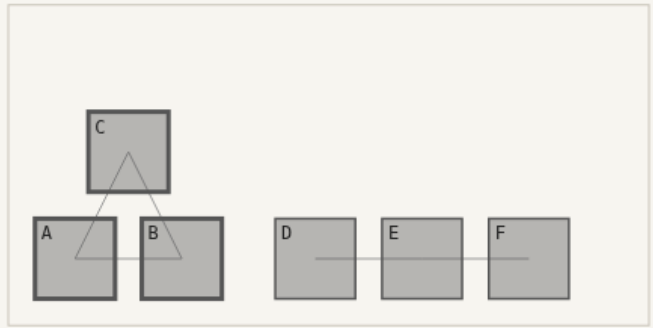
Color attempt

Odd cycle

DOUBLE LELE

Odd cycle

Step 4 / 5 · odd · module01-03-double-lele



Explanation

Triangle A-B-C is an odd cycle.

- odd

```
fail
```

Odd cycle

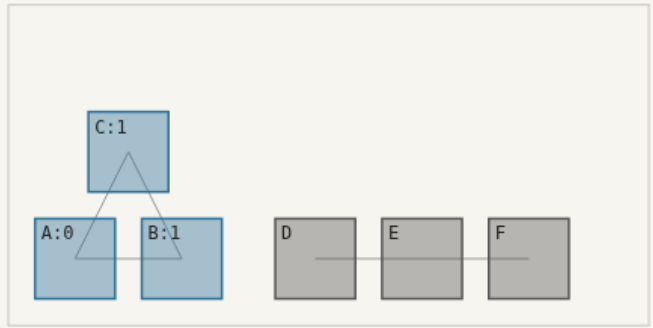
Fail

DOUBLE LELE

Fail

Step 5 / 5 · fail · module01-03-double-lele

DP fail



Explanation

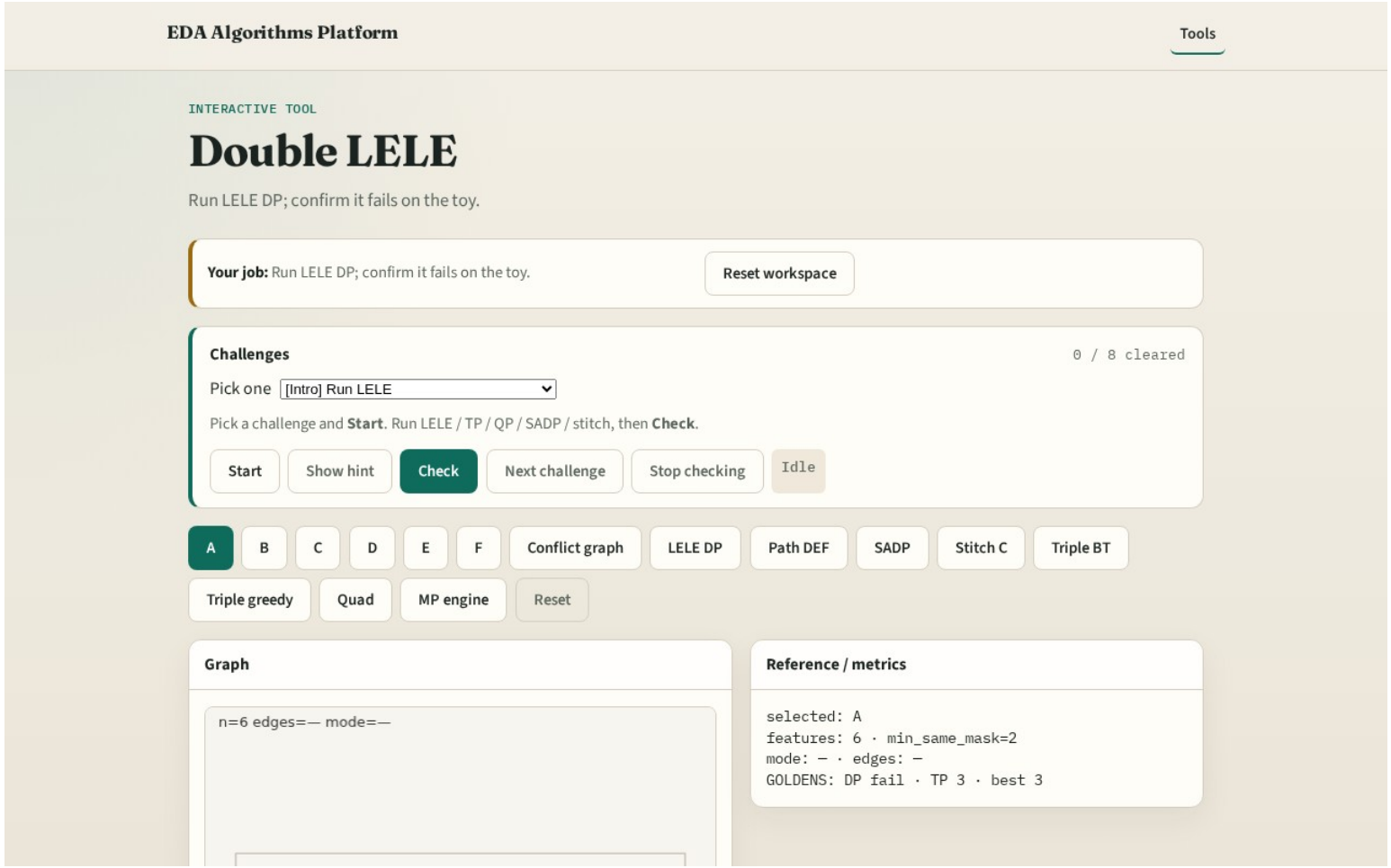
GOLDEN: lele_double ok=False.

- GOLDEN

```
ok=false
```

Fail

Browser lab track



Browser lab starter

Implement track

- Call lele_double(); assert ok is False

Pitfalls

- Assuming DP always works
- Ignoring odd cycles in the full layout

Your turn

- Next: bipartite check on a subgraph